CHAPTER 23

The Digestive System

How the body breaks down food and absorbs nutrients.

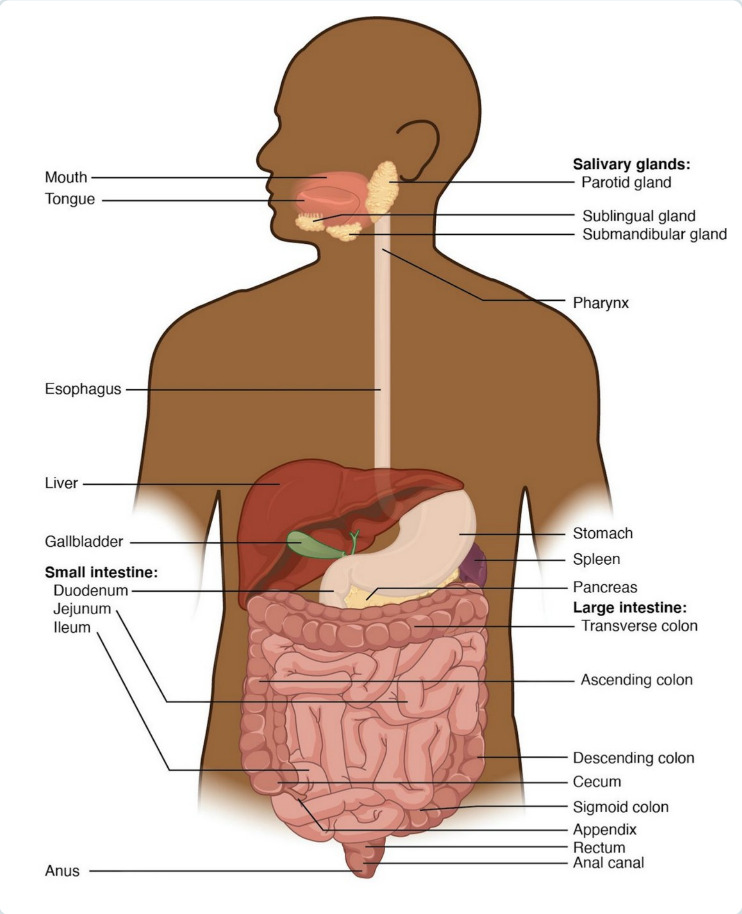
Anatomy & Physiology · Broward-Miami Health Institute

FACULTY EDITION

Learning Objectives

After studying this chapter, students will be able to:

1. List and describe the functional anatomy of the organs and accessory organs of the digestive system
2. Discuss the processes and control of ingestion, propulsion, mechanical digestion, chemical digestion, absorption, and defecation
3. Discuss the roles of the liver, pancreas, and gallbladder in digestion



The organs of the digestive system.

SECTION

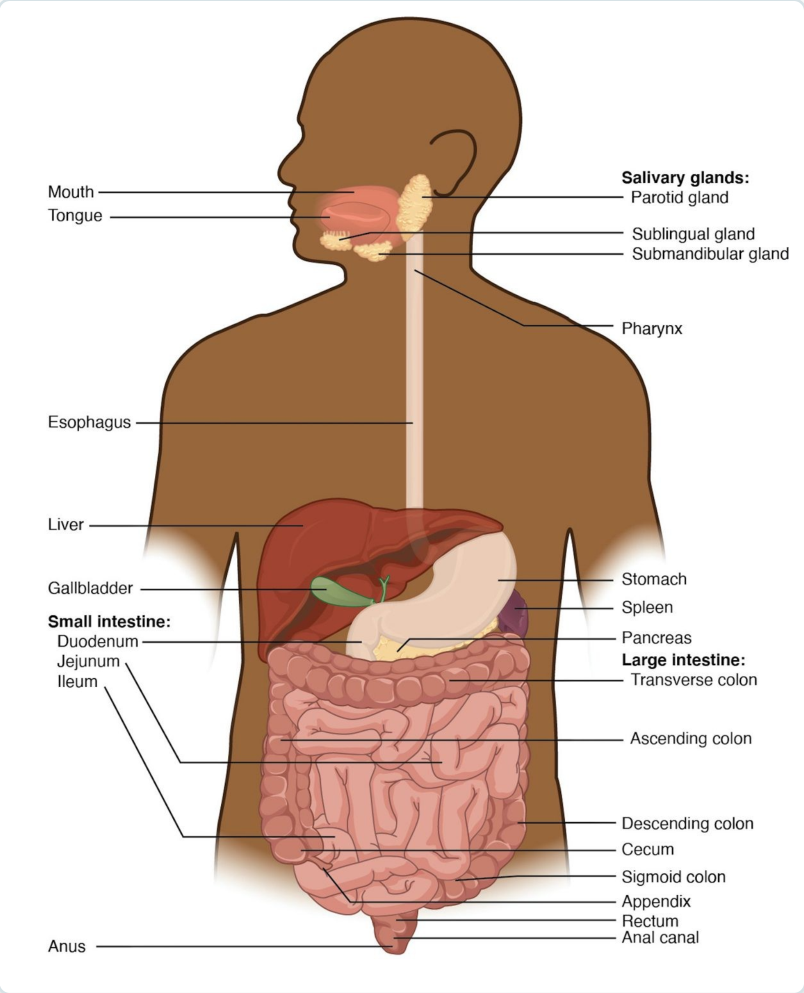
23.1

Overview of the Digestive System

The organs of the digestive tract and its accessory glands.

SECTION 23.1

The Digestive System

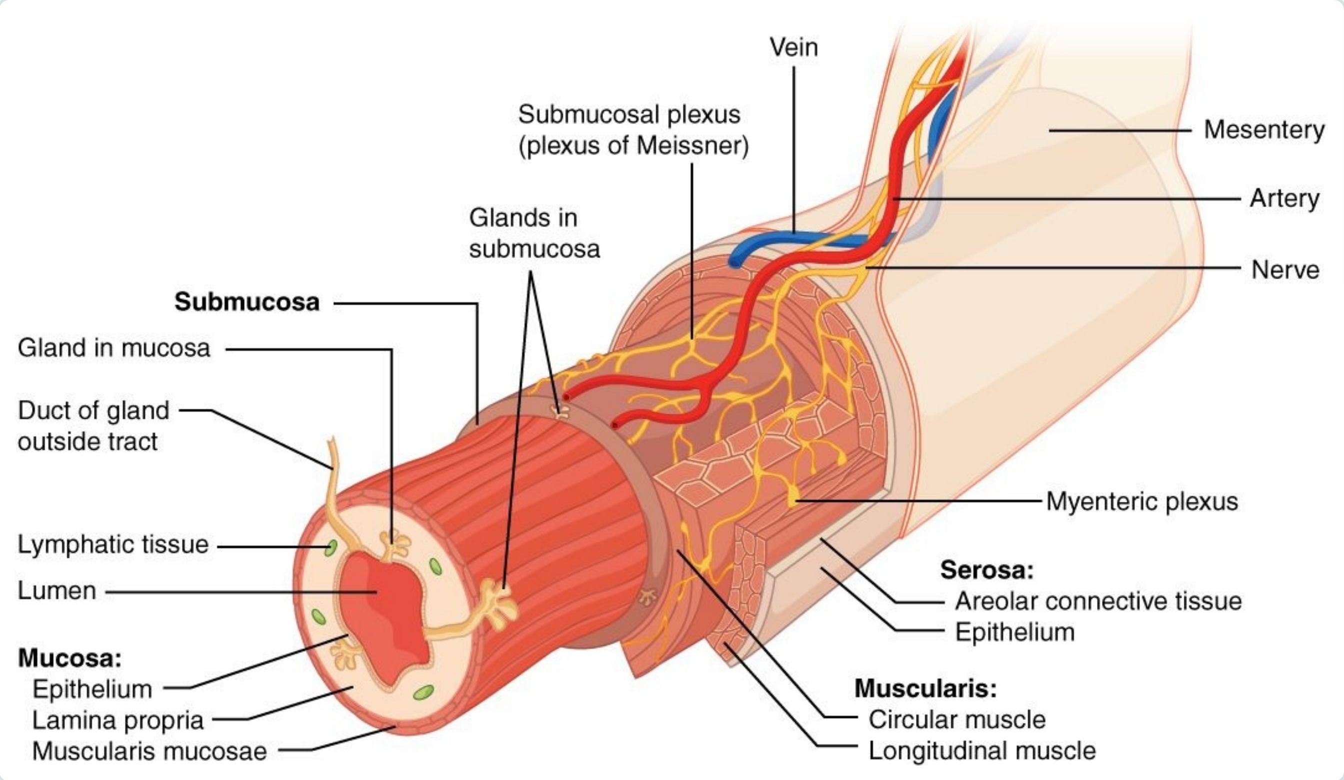


The digestive system

- A long tube from mouth to anus
- The GI tract (alimentary canal)
- Plus accessory organs
- Breaks food into absorbable nutrients

SECTION 23.1

The GI Tract Wall



The layers of the GI tract wall

- Four layers throughout the tract
- Mucosa lines the lumen
- Submucosa carries vessels and nerves
- Muscularis moves food along

23.2

SECTION

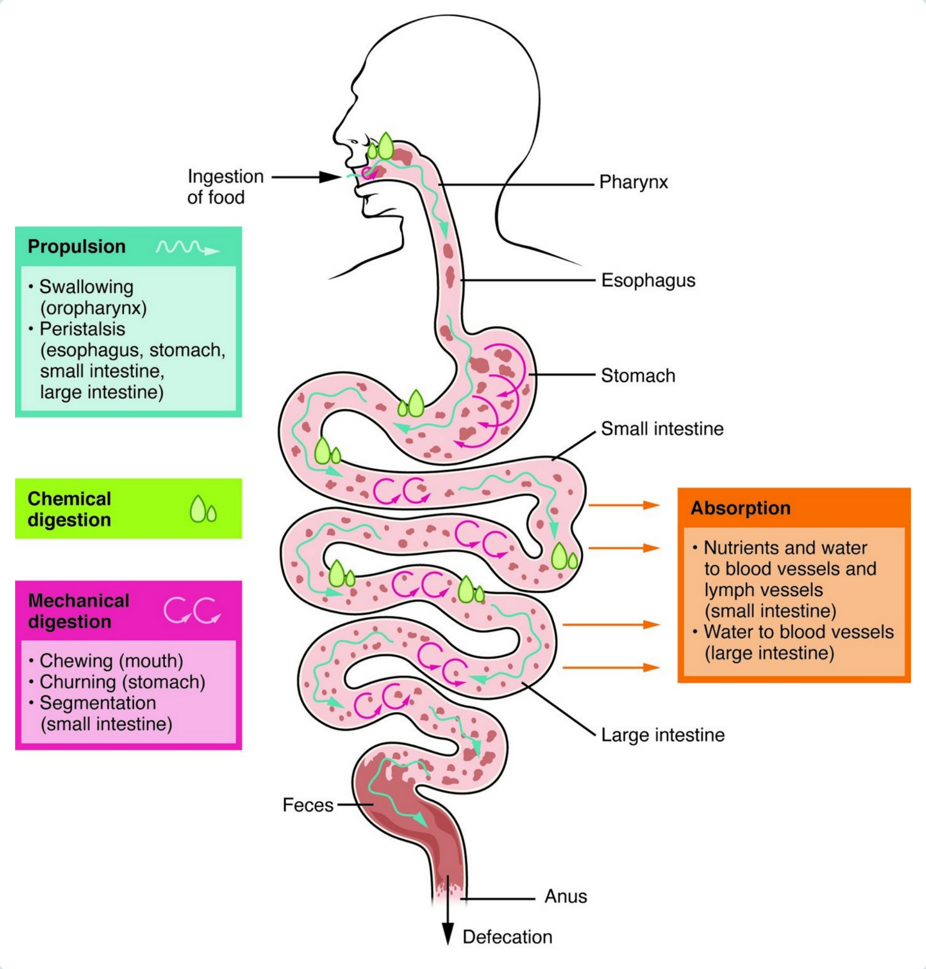
Digestive System Processes and Regulation

The processes of digestion and how they are regulated.

SECTION 23.2

Digestive Processes

- Ingestion takes food in
- Propulsion moves it along
- Mechanical and chemical digestion break it down
- Absorption and defecation finish



The processes of the digestive system

23.3

SECTION

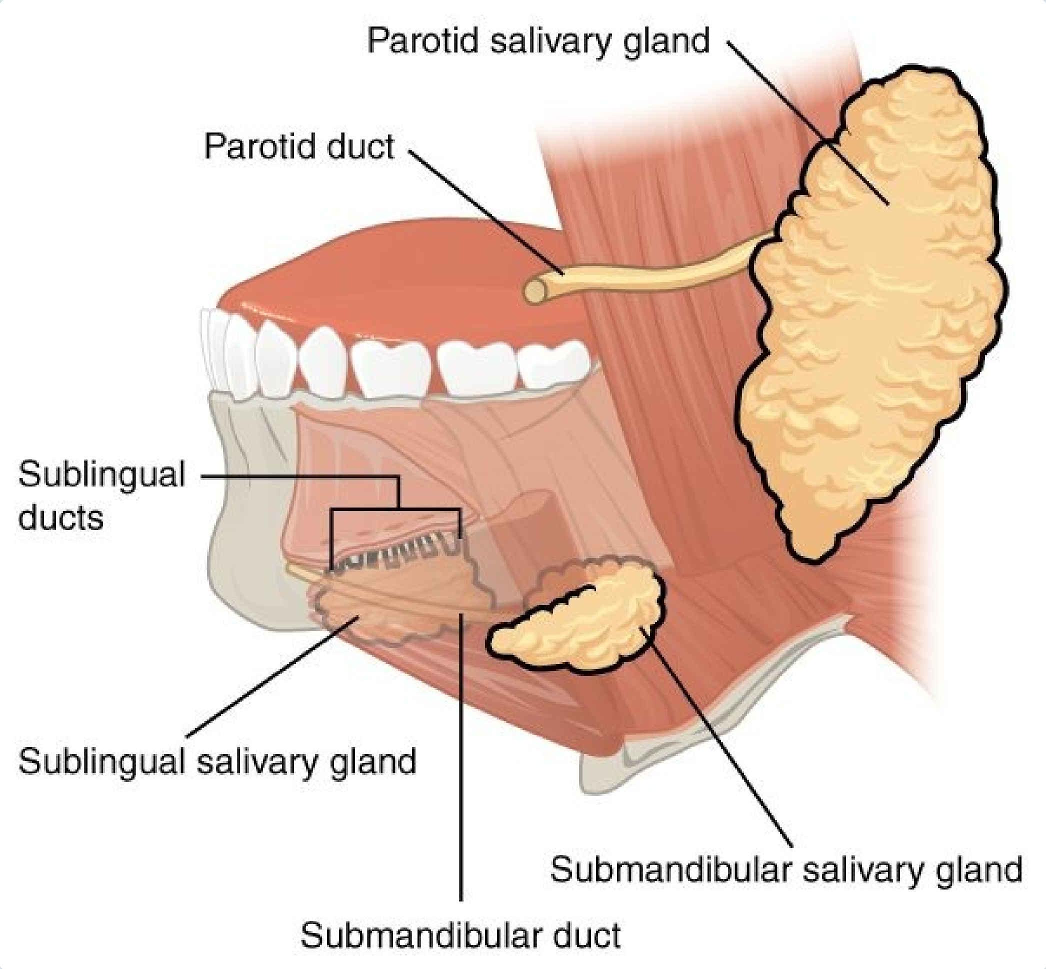
The Mouth, Pharynx, and Esophagus

Where digestion begins — the mouth, pharynx, and esophagus.

SECTION 23.3

The Salivary Glands

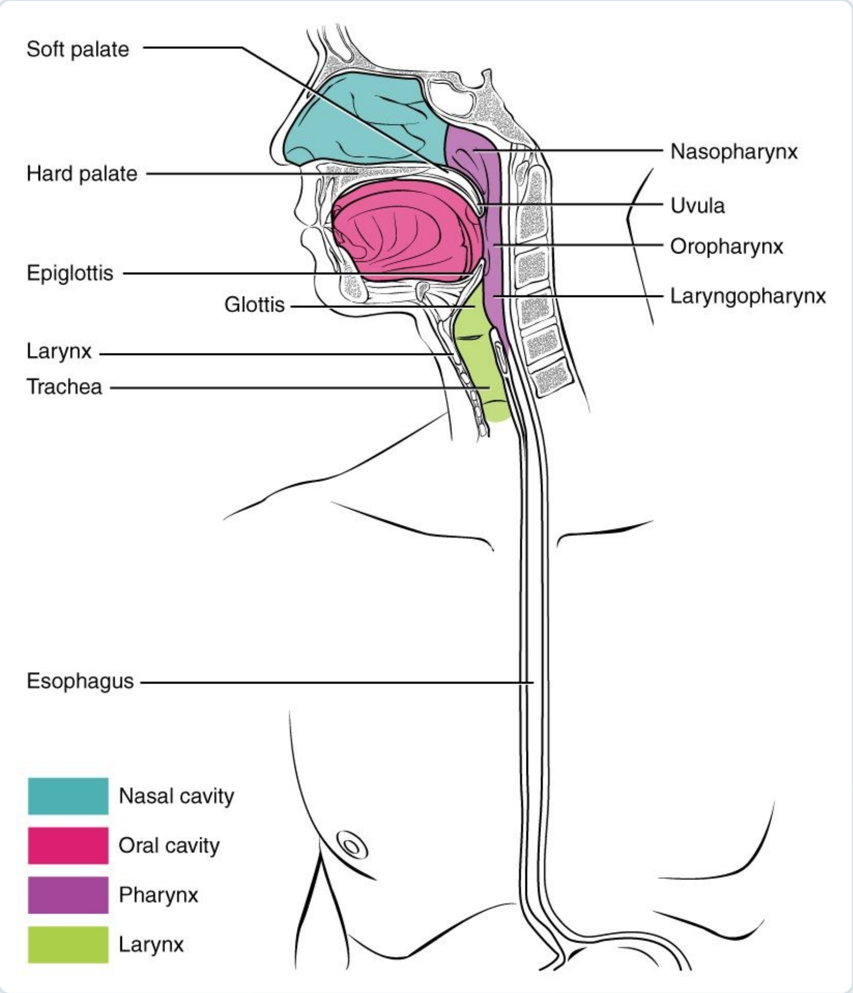
- Three pairs of glands
- Parotid, submandibular, sublingual
- Saliva moistens and begins digestion
- Contains an enzyme for starch



The salivary glands

SECTION 23.3

Swallowing & the Pharynx



The pharynx and swallowing

- The pharynx is shared with the airway
- Swallowing moves the bolus down
- The epiglottis protects the airway
- Peristalsis carries food to the stomach

23.4

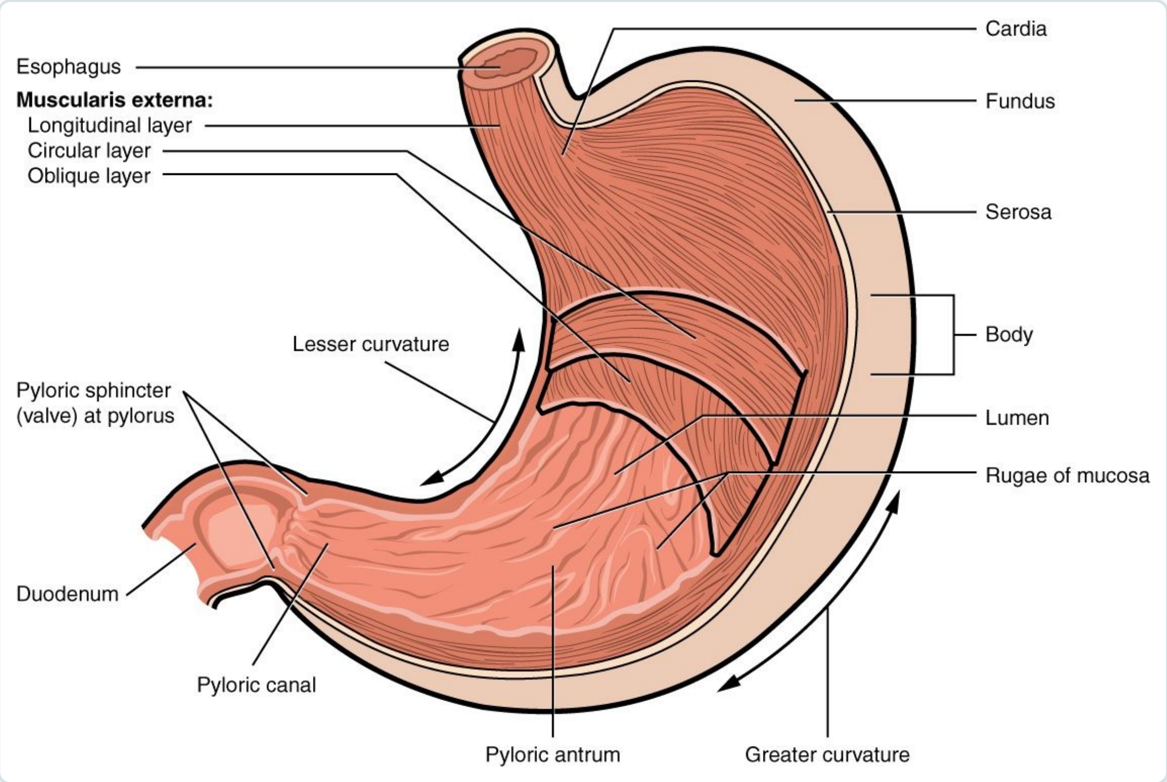
SECTION

The Stomach

The stomach — storing and digesting food into chyme.

SECTION 23.4

The Stomach

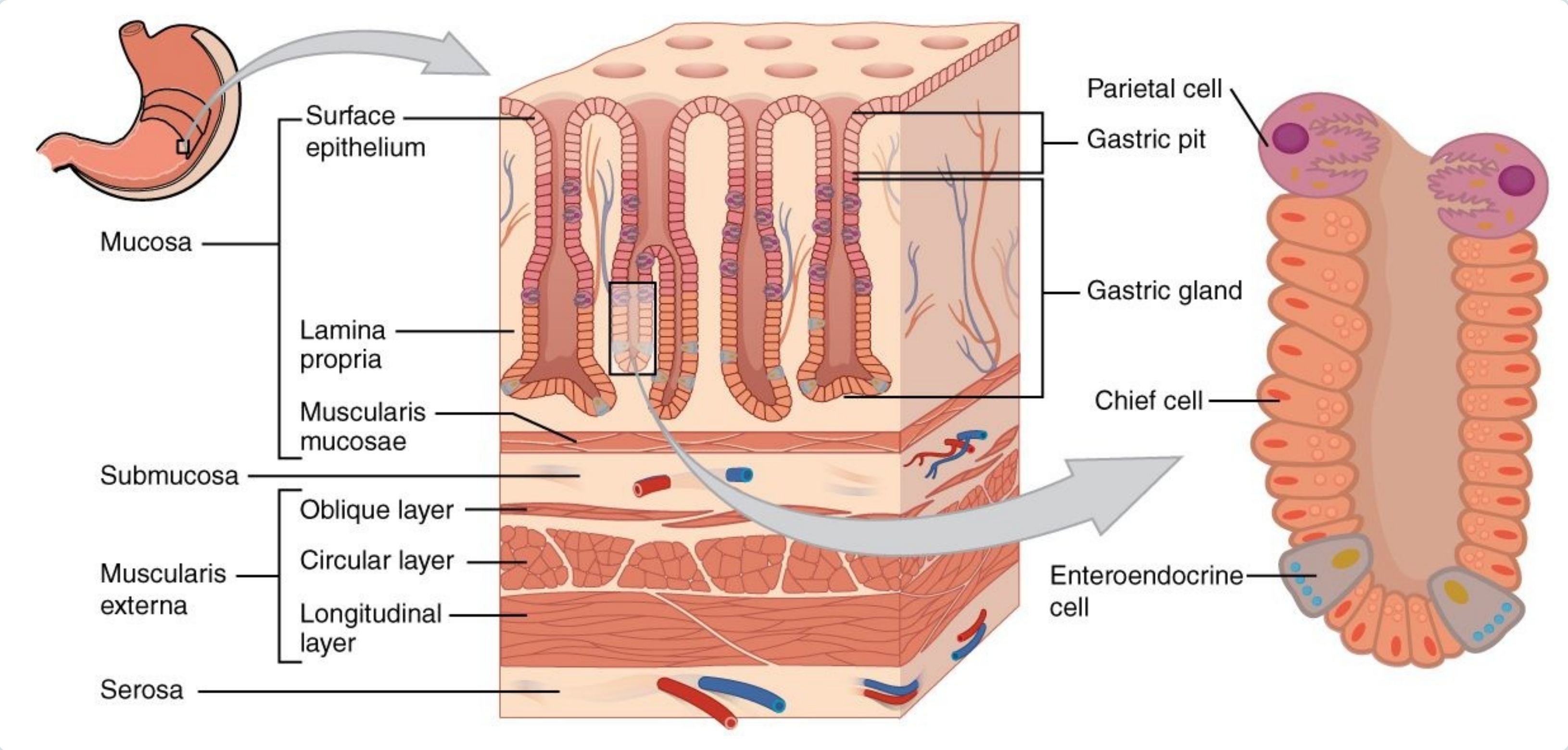


The anatomy of the stomach

- A muscular, J-shaped sac
- Stores and churns food
- Regions: fundus, body, pylorus
- Rugae let it expand

SECTION 23.4

Gastric Glands



The gastric glands

- Gastric pits line the stomach
- Secrete acid and enzymes
- Hydrochloric acid kills microbes
- Pepsin digests protein

23.5

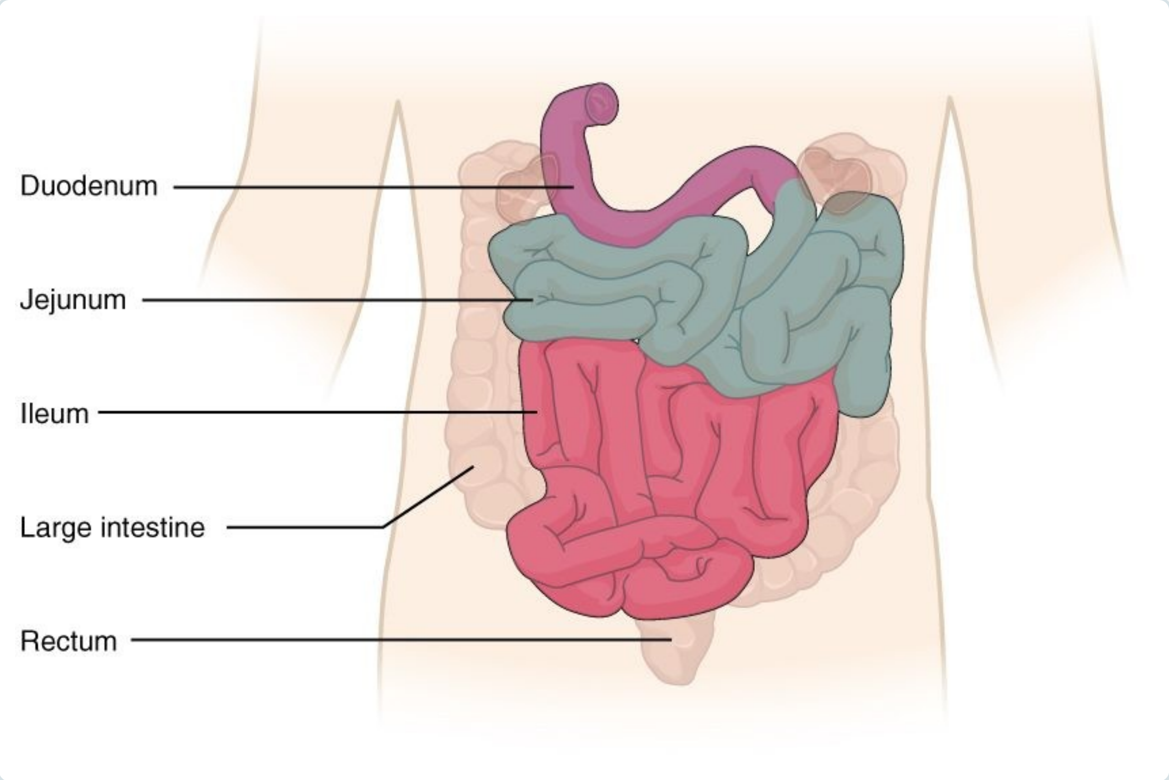
SECTION

The Small and Large Intestines

The intestines — where most digestion and absorption happen.

SECTION 23.5

Small & Large Intestines



The small and large intestines

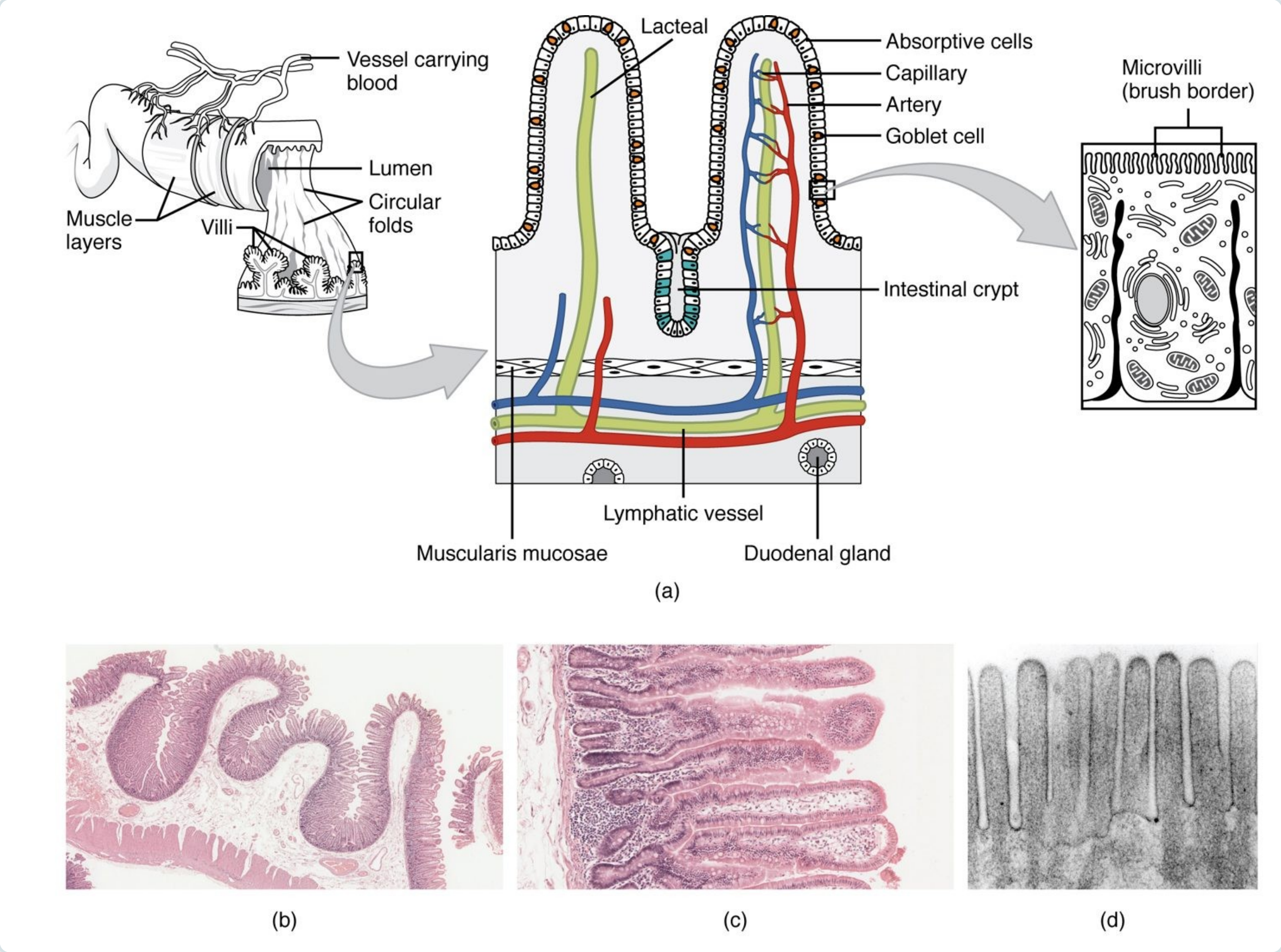
- Small intestine: duodenum, jejunum, ileum
- Most digestion and absorption happen here

- Large intestine absorbs water
- Ends at the rectum and anus

SECTION 23.5

Villi & Absorption

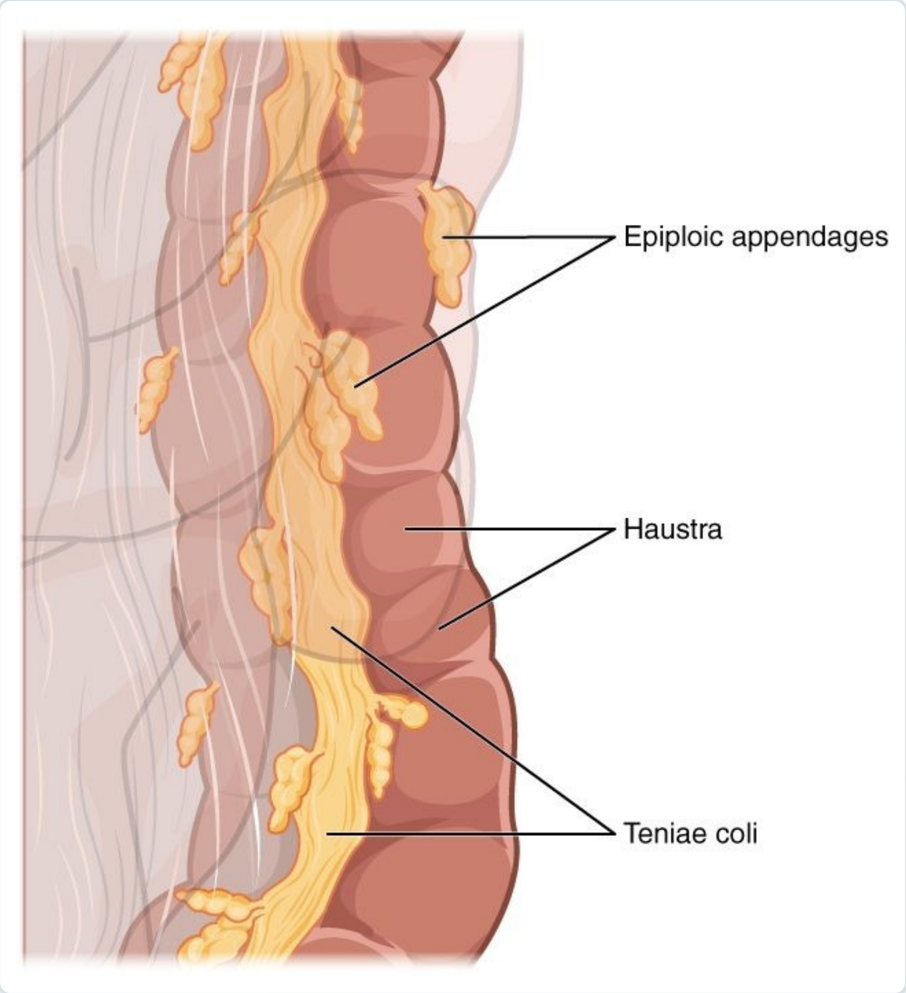
- The small intestine maximizes surface area
- Folds, villi, and microvilli
- A huge area for absorption
- Each villus has vessels and a lacteal



Villi of the small intestine

SECTION 23.5

The Large Intestine



The large intestine

- Frames the abdomen
- Absorbs water and salts
- Pouches (haustra) along its length
- Bacteria make some vitamins

SECTION

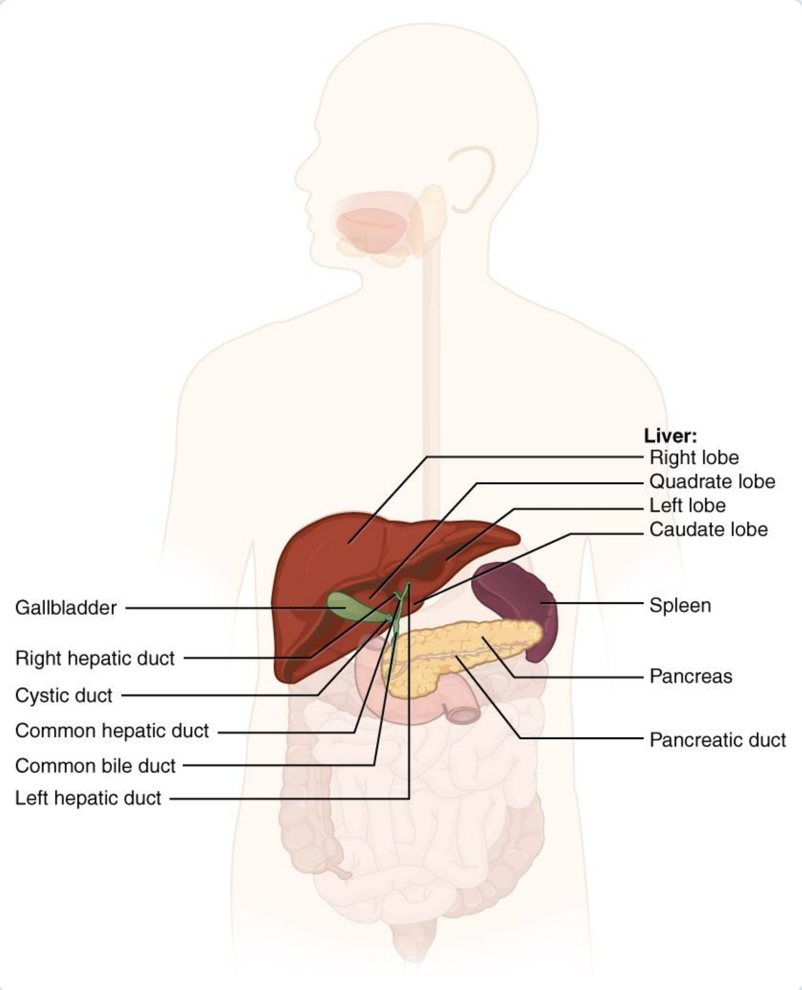
23.6

Accessory Organs in Digestion: The Liver, Pancreas, and Gallbladder

The liver, gallbladder, and pancreas that aid digestion.

SECTION 23.6

The Accessory Organs



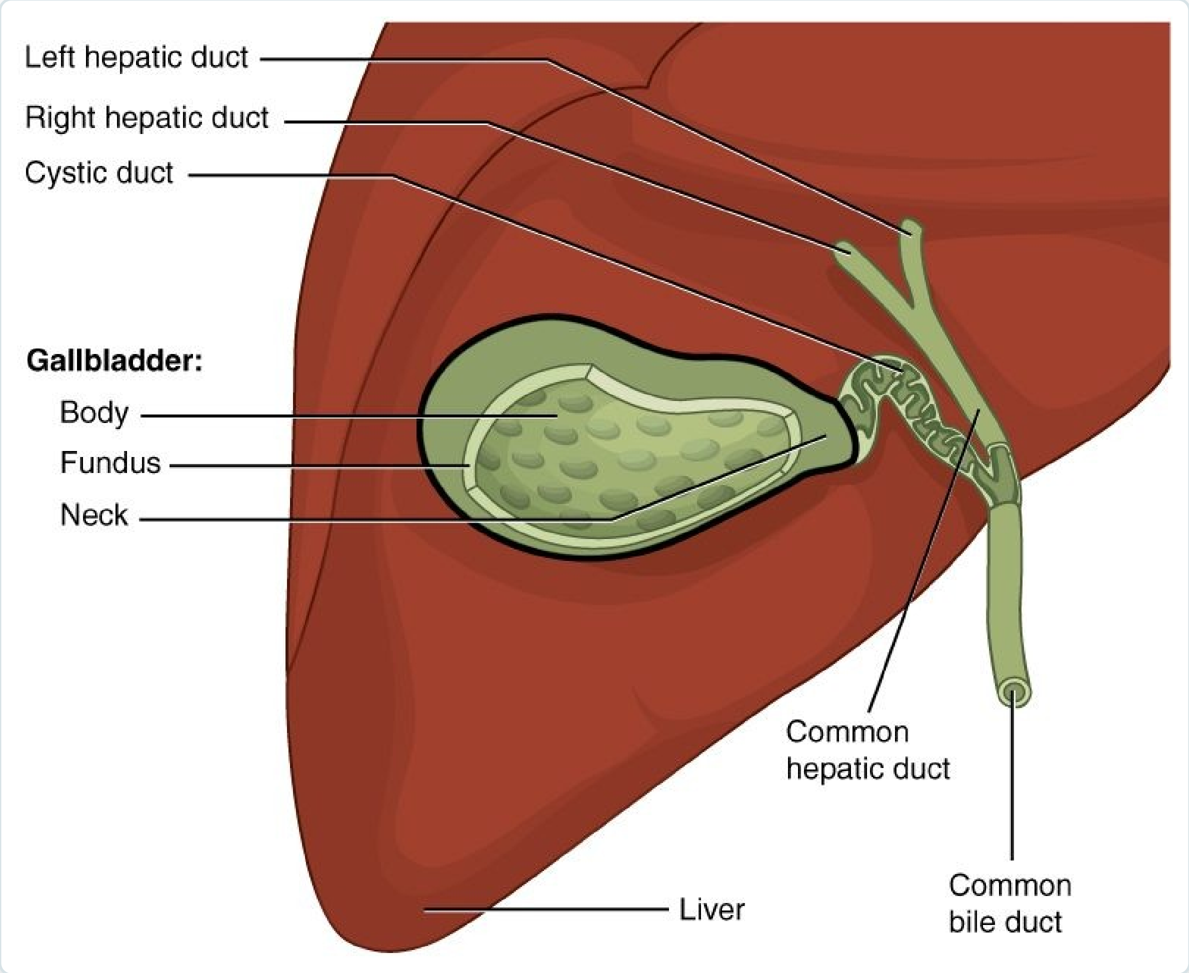
Liver, gallbladder, and pancreas

- Liver, gallbladder, and pancreas
- Secrete into the small intestine
- The liver makes bile
- The pancreas makes digestive enzymes

SECTION 23.6

Gallbladder & Bile Ducts

- The gallbladder stores bile
- Bile emulsifies fats
- Ducts carry bile to the intestine
- Gallstones can block the ducts



The gallbladder and bile ducts

23.7

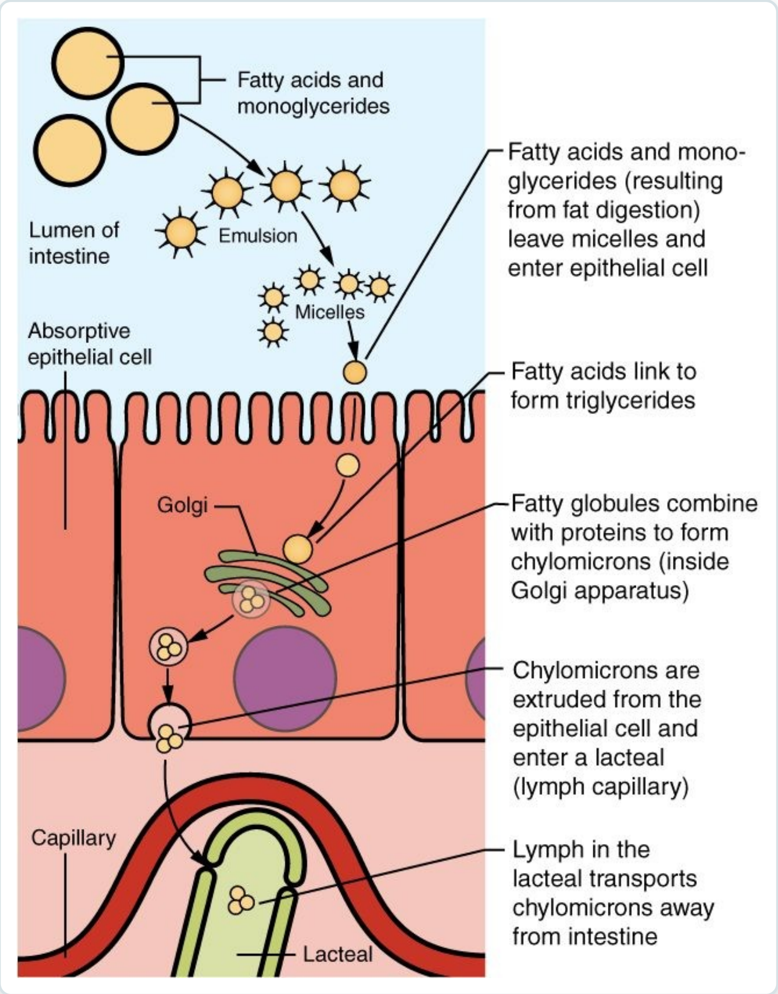
SECTION

Chemical Digestion and Absorption: A Closer Look

How nutrients are chemically broken down and absorbed.

SECTION 23.7

Absorption of Nutrients

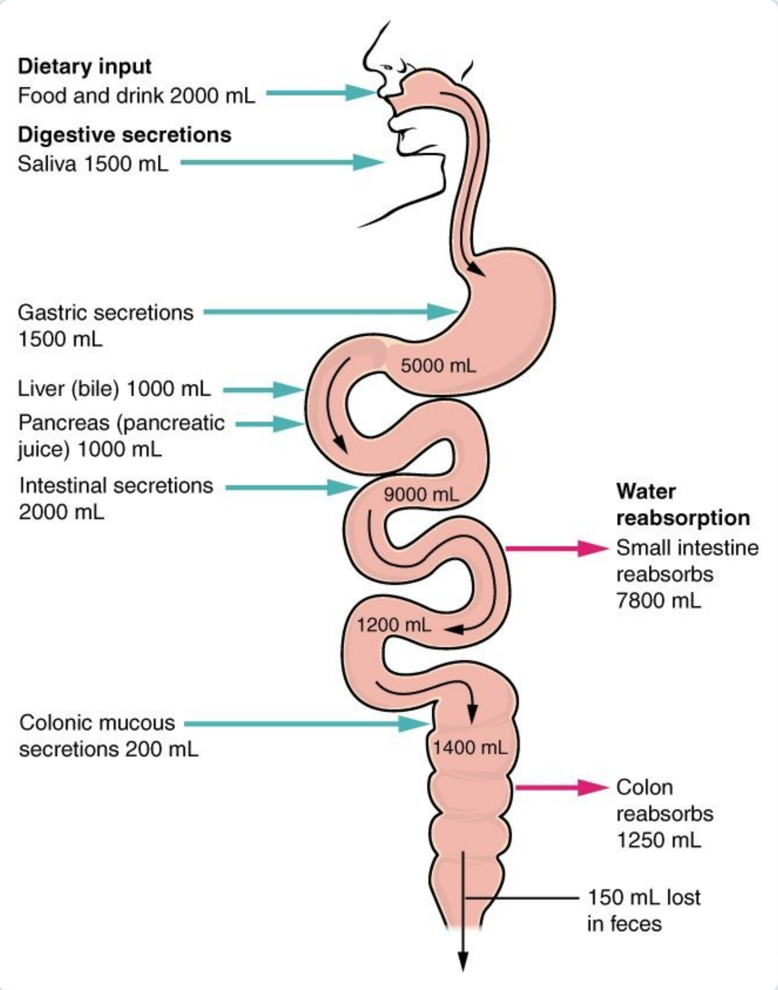


Absorption across the intestinal wall

- Nutrients cross the intestinal lining
- Sugars and amino acids enter the blood
- Fats enter lymphatic lacteals
- Carried to the body's cells

SECTION 23.7

Fluid Balance in Digestion



Water balance in the digestive tract

- The gut handles large fluid volumes
- Most is reabsorbed, not lost
- The small intestine absorbs the most
- Diarrhea risks dehydration

CHAPTER 23

Key Takeaways

- The digestive system is a long tract plus accessory organs that process food.
- Digestion involves mechanical and chemical breakdown, then absorption.
- The mouth and stomach begin digestion; the small intestine completes most of it.
- The liver, gallbladder, and pancreas supply bile and enzymes for digestion.
- The large intestine absorbs water and forms stool for elimination.

CLINICAL CONNECTIONS

Why the Digestive System Matters in Practical Nursing

Dysphagia & Aspiration

Trouble swallowing risks food entering the airway.

Peptic Ulcers

Acid damage to the stomach or duodenal lining.

Gallstones

Hardened bile can block ducts and cause severe pain.

Diarrhea & Dehydration

Lost intestinal fluid can quickly unbalance the body.

Nutrition

Digestion and absorption underlie all nutritional care.

Liver Function

The liver is central to digestion, metabolism, and detoxification.